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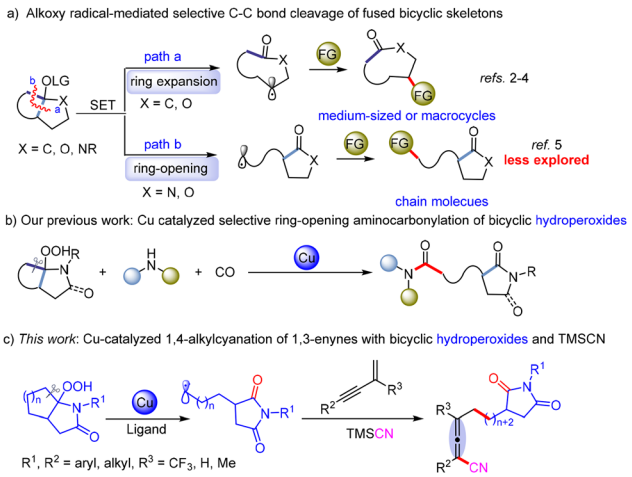
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Copper-catalyzed 1,4-alkylcyanation of CF_3-containing 1,3-enynes: alkoxy radical-mediated selective cleavage of fused azabicycles

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A copper-catalyzed 1,4-alkylcyanation of CF_3-containing 1,3-enynes with bicyclic hydroperoxides and TMSCN is described. In this cascade, the alkyl source originates from selective cleavage of a C–C bond in the fused bicyclic skeletons, enabling the successful introduction of a succinimide fragment into tetrasubstituted allenes. This reaction is characterized by mild conditions, broad substrate scope, excellent functional group tolerance, and ease of large-scale synthesis and product derivatization.

In recent years, radical-mediated C–C bond cleavage of unstrained aliphatic carbocycles has emerged as a powerful platform for the formation of various chemical bonds. Significant progress has been made in the alkoxy-radical mediated ring-opening of monocarbocycles, which enables the successful introduction of ketoalkyl fragments into various compounds.[1] In contrast, the alkoxy radical-induced C–C bond cleavage of fused bicyclic skeletons remains less developed, even though there were pioneering reports in the 1980s (Scheme 1a).[2] In particular, the selective cleavage of fused bicyclic skeletons remains a significant yet relatively less explored area. Recently, Zuo's group reported an elegant photoinduced Ce-catalyzed aerobic oxidative ring expansion of hemiketal alcohols (tautomer mixture) to carbonyl-containing macrolatones *via* a ligand-to-metal charge transfer (LMCT) strategy.[3] Our group has developed several abundant metal-catalyzed ring expansion/cross-coupling cascades of hemiketal hydroperoxides, yielding a range of site-specific functionalized macrolactones.[4] Remarkably, by adjusting the substituents on the bicyclic skeletons,[5a] we achieved an unexpected Cu-catalyzed ring-opening/aminocarbonylation cascade, yielding amides that contain a γ-lactam or succinimide fragment instead (Scheme 1b).[5b] DFT calculations revealed that the regioselective ring-opening is thermodynamically and kinetically favorable, due to the formation of a stable γ-lactam or succinimide fragment and an unstrained alkyl radical. These intriguing findings motivated us to explore new radical alkylation reactions, with the aim of incorporating these functionalized alkyl moieties into diverse molecules.



Scheme 1 Alkoxy radical-mediated selective cleavage of fused bicycles (a–c).

Radical-mediated multicomponent reactions represent an efficient strategy for the rapid assembly of complex molecules from simple starting materials, characterized by mild reaction conditions, high step-economy and superior bond formation efficiency. In this field, 1,3-enynes have been extensively used as versatile linkage bridges.[6] A variety of three-component functionalization reactions of 1,3-enynes have been developed including 1,2-functionalization,[7] 1,4-functionalization,[8] and 3,4-functionalization.[9] Among these, the radical-mediated 1,4-difunctionalization of 1,3-enynes has gradually emerged as an important and powerful strategy for the synthesis of polysubstituted allenes.[10] In particular, the copper-catalyzed 1,4-alkylcyanation of 1,3-enynes provides an efficient method for the rapid synthesis of CN-functionalized tetrasubstituted allenes. A range of alkyl radical precursors, including alkyl

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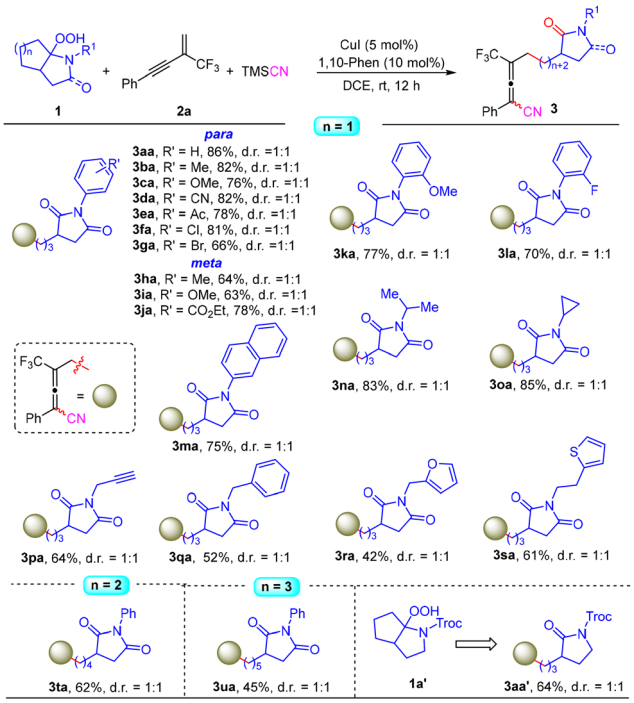
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diacyl hydroperoxides,[11] cyclobutanone oxime esters,[12] alkyl *N*-hydroxyphthalimide esters,[13] and others[14] have been successfully applied for this cascade. In addition, Wu's group and our group have reported the four-component 1,4-alkylcarbonylation reaction of 1,3-enynes, providing a straightforward approach to complex allenyl ketones.[15] Given the significance and versatile reactivity of allenes, we attempt to integrate our selective C–C bond cleavage, ring-opening findings with the 1,4-alkylcyanation of 1,3-enynes (Scheme 1c). Herein, we disclose a copper-catalyzed 1,4-alkylcyanation cascade of CF_3-contaning 1,3-enynes using fused azabicyclic hydroperoxides and TMSCN. This protocol provides efficient access to succinimide-containing tetrasubstituted allenes, which are otherwise inaccessible by conventional methods.

To achieve this goal, we selected the reaction of fused bicyclic hydroperoxide **1a**, CF_3-containing 1,3-enyne **2a** and TMSCN as a model reaction to identify the optimal reaction conditions under copper catalysis (for details, see Table S1 in the SI). We found that the three-component reaction proceeded successfully in the presence of CuI (5 mol%), and 1,10-phenanthroline (1,10-Phen, **L1**, 10 mol%) in DCE at room temperature, yielding the desired product **3aa** in a 74% yield after 12 h. Other Cu(I) and Cu(II) salts were then examined for this transformation, and CuI still showed the best catalytic activity (entries 1–6). Solvent screening indicated that THF also afforded a satisfactory yield of **3aa**, yet DCE remained the optimal solvent (entries 7–10). Sequentially, various bidentate and tridentate nitrogen ligands, including 5,6-dimethyl-1,10-phen (**L2**), 4,4′-di-*tert*-butyl-2,2′-bipyridine (dtbpy, **L3**) and 2,2′:6′,2″-terpyridine (**L4**) were tested. Among these, the tridentate ligand was totally ineffective, while the simple 1,10-Phen remained the most efficient (entries 11–13). Increasing the amounts of **1a** and TMSCN to 2.0 equiv. further enhanced the yield of **3aa** to 86% (entry 14). Finally, control experiments confirmed that both the Cu catalyst and the ligand were essential for the success of this transformation (not shown).

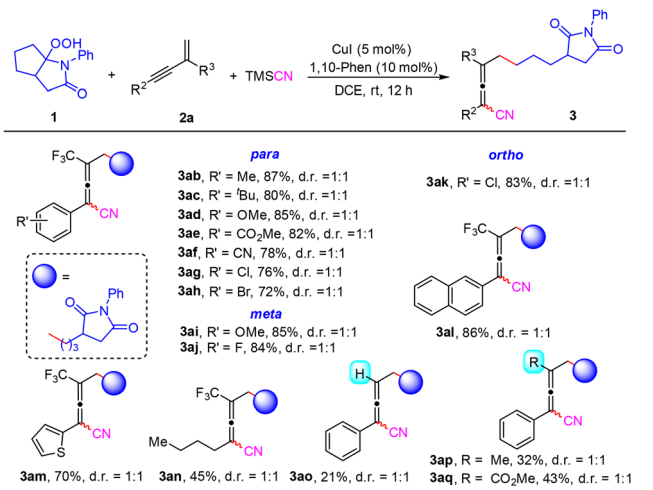
With the optimal conditions in hand, the substrate scope of fused bicyclic hydroperoxides **1** was first evaluated using 1,3-enyne **2a** and TMSCN as coupling partners (Scheme 2). It was found that *N*-aryl hydroperoxides bearing electron-donating or -withdrawing groups on the aromatic ring (at the para, meta or ortho positions) participated efficiently in this transformation, yielding the desired products **3ba–3la** in good yields. Notably, a wide range of functional groups, including ether (**3ca**, **3ka**), cyano (**3da**), acetyl (**3ea**), halogen (**3ga**), and ester (**3ja**) groups were well tolerated in this reaction. The *N*-(2-naphthyl) substrate also reacted smoothly, producing the allene **3ma** in a 75% yield. Besides *N*-aryl hydroperoxides, *N*-alkyl and *N*-benzyl substituted hydroperoxides were also found to be effective in this reaction, yielding the desired products **3na–3sa** in moderate to good yields. Functional groups such as cyclopropyl (**3oa**), alkynyl (**3pa**), furyl (**3ra**) and thienyl (**3sa**) groups were all compatible with this reaction. Furthermore, the reaction was not limited to bicyclo[3.3.0] hydroperoxides; bicyclo[4.3.0] and [5.3.0] hydroperoxides also yielded the expected products **3ta** and **3ua**, albeit with somewhat lower yields. Notably, when bicyclic hydroperoxide **1a**′ was subjected to the reaction, a



Scheme 2 Scope of bicyclic hydroperoxides. Reaction conditions: **1** (0.40 mmol, 2.0 equiv.), **2a** (0.20 mmol, 1.0 equiv.), CuI (5 mol %), 1,10-Phen (10 mol %) and TMSCN (0.40 mmol, 2.0 equiv.) in DCE (1.5 mL) at room temperature for 12 h under N_2, isolated yields. d.r. ratios were determined by ^{13}C NMR spectra or HPLC analysis.

regioselective C–C bond cleavage occurred, producing the ring-opening product **3aa**′ in a 64% yield. The result is different from the *oxo*-bicyclic analogues in our previous report.[14]

Then, the generality and limitations of 1,3-enynes were investigated using **1a** as the alkylation reagent (Scheme 3). 1,3-Enynes bearing aryl, heteroaryl and alkyl substituents in

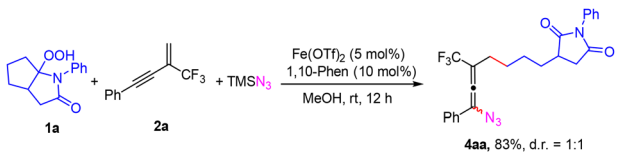


Scheme 3 Substrate scope of 1,3-enynes. Reaction conditions: **1** (0.40 mmol, 2.0 equiv.), **2a** (0.20 mmol, 1.0 equiv.), CuI (5 mol %), 1,10-Phen (10 mol %) and TMSCN (0.40 mmol, 2.0 equiv.) in DCE (1.5 mL) at room temperature for 12 h under N_2, isolated yields. d.r. ratios were determined by ^{13}C NMR spectra or HPLC analysis.

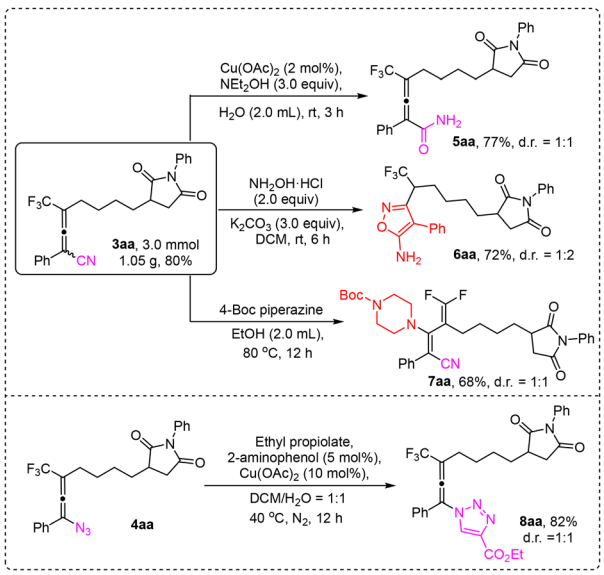
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the alkyne moiety reacted efficiently with **1a** and TMSCN, furnishing the desired products **3ab–3an** in good yields. Both electronic and steric effects of substituents on the aromatic ring did not show a significant impact on the reaction efficiency. Additionally, synthetically useful functional groups such as ester (**3ae**), cyano (**3af**) and bromo (**3ah**) were all compatible with this catalytic system. When the trifluoromethyl group on the 1,3-enyne was replaced by a hydrogen atom, a methyl group and a methyl ester group, the corresponding products **3ao**, **3ap** and **3aq** were successfully obtained, but with noticeably reduced yields. This decrease in yield might be due to a polarity mismatch. The alkyl radicals generated by the carbon–carbon bond cleavage are nucleophilic, and thus, electron-deficient CF_3-containing 1,3-enynes exhibited better reactivity than electron-rich 1,3-enynes. Remarkably, when $TMSN_3$ was used instead of TMSCN as the nucleophile, the 1,4-alkylazidation of CF_3-containing 1,3-enyne proceeded smoothly under iron catalysis (Scheme 4), yielding the expected product **4aa** in a satisfactory yield (83%).

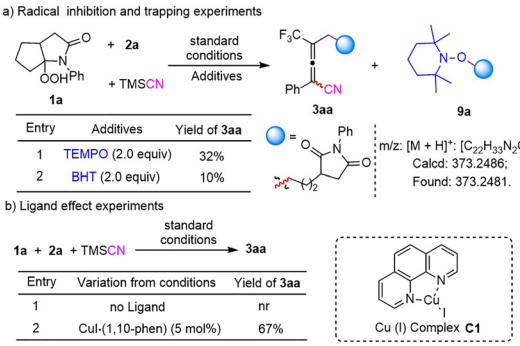
To further demonstrate the application potential of this reaction, a gram-scale amplification experiment was carried out (Scheme 5). When the model reaction was scaled up to 3.0 mmol, the target product **3aa** could be obtained in an 80% yield (1.05 g). Subsequently, various derivatizations of **3aa** were explored. The hydrolysis of the cyano group in allene **3aa** was successfully achieved using a copper-catalyzed system, yielding the 2,3-dienamide **5aa** in a 77% yield. Under alkaline conditions, allene **3aa** could undergo a nucleophilic cyclization reaction with



Scheme 4 1,4-Alkylazidation of CF_3-containing 1,3-enyne.



Scheme 5 Gram-scale synthesis and derivatizations of the products.

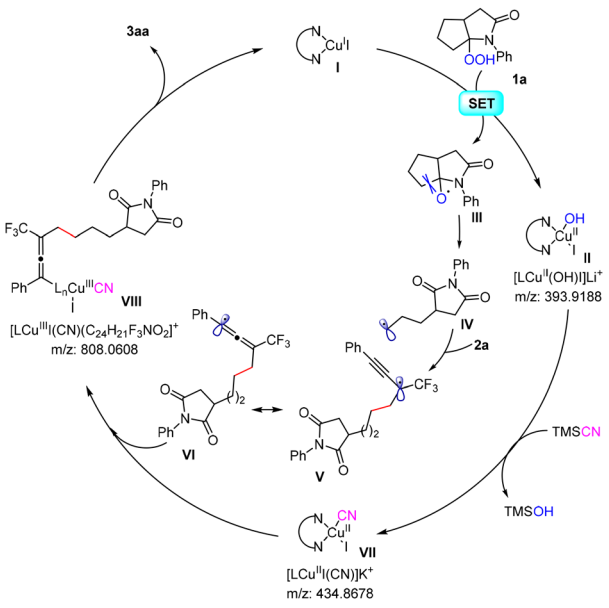


Scheme 6 Mechanism studies (a and b).

hydroxylamine hydrochloride, affording the isoxazole **6aa** in a 72% yield. Moreover, using 4-Boc piperazine as the nucleophile, the *gem*-difluorodiene **7aa** was obtained in a satisfactory yield through a nucleophilic addition followed by a defluorination process. Finally, the derivatization of azide-containing allene **4aa** was explored. Satisfactorily, it could undergo a copper-catalyzed [3+2] cycloaddition click reaction with ethyl propiolate efficiently, yielding the triazole-substituted allene **8aa** in an 82% yield.

To shed light on the reaction mechanism, radical inhibition and trapping experiments were first performed (Scheme 6a). When 2.0 equiv. of TEMPO or BHT, well known radical scavengers, were added to the model reaction, the yield of **3aa** was significantly reduced. Additionally, the alkyl-TEMPO adduct **9a** could be detected by HRMS. These results indicated that a radical pathway is likely involved in this transformation. Furthermore, control experiments indicate that the Cu(I) complex **C1** is a key active species in the catalytic cycle of this reaction (Scheme 6b).

Based on the above experimental results and literature,[11–13] a possible reaction mechanism was proposed. As shown in Scheme 7, the reaction begins with the single-electron



Scheme 7 Proposed mechanism.

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reduction of the fused bicyclic hydroperoxide **1a** by the Cu(I) complex **I** formed *in situ*, yielding the radical intermediate **III** and the Cu(II) species **II**. Subsequently, the alkoxy radical **III** undergoes a selective β-scission to give the ring-opening, relatively stable alkyl radical **IV**. Intermediate **IV** is captured by the CF_3-1,3-enyne **2a** to form a propargyl radical species **V**, which resonates to produce the allenyl radical **VI**. Meanwhile, the Cu(II) species **II** undergoes a ligand exchange with TMSCN to yield the key intermediate **VII**. The interaction of intermediates **VI** and **VII** forms a Cu(III) species **VIII**. Finally, reductive elimination of Cu(III) species **VIII** gives the target product **3aa** and regenerates the Cu(I) complex **I**. Fortunately, all relevant copper complexes can be detected by HRMS.

In summary, we have developed a copper-catalyzed 1,4-alkylcyanation of 1,3-enynes with bicyclic hydroperoxides and TMSCN. In this cascade, the bicyclic hydroperoxides serve as a functionalized alkyl source *via* an alkoxy radical-mediated selective β-scission, enabling the introduction of a succinimide fragment into polysubstituted allenes. This method has significant advantages, including mild reaction conditions, excellent functional group tolerance, and ease of large-scale synthesis and product derivatization. Additionally, this cascade can be extended to the 1,4-alkylazidation of 1,3-enynes. This work not only enriches the diverse synthesis of functionalized allenes but also provides a good example of selective C–C bond cleavage in bicyclic skeletons.

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Conflicts of interest

There are no conflicts to declare.

Data availability

The data supporting this article have been included as part of the SI. All experimental data, spectra, and detailed experimental procedures are available in the SI. See DOI: **https://doi.org/10.1039/d5cc03900b**

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